

Name: Abhishek Hegde



CAMBRIDGE INFOTECH

Transformers & AI Application Architecture

60-Question Assessment

Parts A–D: 40 concept questions (10 each) • Parts E–H: 20 scenario-based questions (5 each) •
multiple choice

Part A — AI Application Architecture

1. What are the five functional layers of a production AI application pipeline, in order?
 - A. Input, Retrieval, Orchestration, Observability, Safety
 - B. Input, Orchestration, Retrieval & Tools, Safety & Quality, Observability
 - C. Orchestration, Input, Safety, Retrieval, Observability
 - D. Safety, Input, Orchestration, Observability, Retrieval
2. What is the primary purpose of the Input Layer?
 - A. To log cost and latency
 - B. To validate and contextualize a query before it consumes compute
 - C. To route the query to the cheapest model
 - D. To score the output for accuracy
3. What does the Agent Orchestrator do?
 - A. Filters toxic content
 - B. Coordinates the overall workflow and decides which layer runs next
 - C. Stores long-term user memory
 - D. Tracks token spend per request
4. What does the Multi-Model Router decide?
 - A. Which model handles a given request, based on task complexity and cost
 - B. Whether a query passes safety checks
 - C. How many tokens to log
 - D. Whether to retry a failed response
5. What is the RAG pipeline used for in the Retrieval & Tools layer?
 - A. Scoring output tone
 - B. Finding relevant passages from company documents and injecting them into the prompt
 - C. Blocking jailbreak attempts
 - D. Measuring latency
6. What are the three stages of the Safety & Quality layer?
 - A. Input validation, routing, logging
 - B. Output guardrails, evaluation framework, quality check gate
 - C. Tokenization, embedding, attention
 - D. Cost tracking, latency tracking, eval dashboard
7. What happens when a response fails the Quality Check?
 - A. It's shown to the user with a warning
 - B. It's sent back to the Orchestrator for retry / refine
 - C. It's discarded permanently

D. It's logged and ignored

8. What three things does the Observability layer track?

- A. Accuracy, relevance, tone
- B. Cost, latency/performance, eval metrics
- C. Prompt injection, toxicity, scope
- D. Query, key, value

9. Why is input validation described as "the cheapest place to stop a bad request"?

- A. It happens after the model call, saving storage
- B. It runs before any model call, avoiding the cost of a full model call
- C. It only checks formatting
- D. It is done manually by a human

10. According to the deck's takeaways, why doesn't every query need the biggest model?

- A. Big models are always cheaper
- B. Routing simple tasks to cheap/fast models and hard reasoning to frontier models controls cost without hurting quality
- C. Small models are more accurate
- D. Users prefer faster wrong answers

Part B — A Field Guide to Transformers

11. Fundamentally, what does a GPT-based language model do on repeat?

- A. Translate text into vectors permanently
- B. Predict the next word, feed it back in, and predict again
- C. Search a database for the answer
- D. Classify the entire input at once

12. What is a word embedding?

- A. A list of numbers (vector) capturing a word's meaning
- B. A dictionary definition
- C. A grammar rule
- D. A binary flag for word type

13. How many dimensions per word does BERT use for its embeddings?

- A. 96
- B. 768
- C. 12,288
- D. 30,000

14. What is the key limitation of static embeddings like Word2Vec or GloVe?

- A. They take too long to compute

B. Every word gets one fixed vector regardless of context, so the same word can't capture multiple meanings

C. They can't be used for English

D. They are too large to store

15. In the full Transformer, what does the encoder produce?

A. The final softmax prediction

B. One contextual embedding per input token

C. A single fixed-length vector for the whole sentence

D. The tokenizer vocabulary

16. Which models use only the encoder half and only the decoder half of the Transformer, respectively?

A. GPT uses encoder only, BERT uses decoder only

B. BERT uses encoder only, GPT uses decoder only

C. Both use encoder only

D. Both use decoder only

17. What are the three steps for getting words into a Transformer model?

A. Tokenize, static embed, add position

B. Train, infer, evaluate

C. Query, key, value

D. Encode, decode, generate

18. What is the key difference between training and inference?

A. Training predicts word by word; inference adjusts weights

B. Training adjusts model weights via back-propagation; inference uses the fixed, learned weights to predict

C. They are the same process

D. Inference happens before training

19. In the Query/Key/Value library analogy, what does the Key represent?

A. What you're looking for

B. How each candidate (book) describes itself

C. The actual information you walk away with

D. The final prediction

20. What does the softmax function do in $\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_K})V$?

A. Multiplies query and key vectors

B. Converts raw relevance scores into percentages that sum to 100%

C. Adds positional information

D. Compresses the vector dimensions

Part C — Understanding Transformers (The 7-Step Pipeline)

21. What are the seven ideas that explain what a transformer does to a sentence?

- A. Tokenization, static embeddings, positional embeddings, Q/K/V attention, multi-head attention, feed-forward refinement, stacking N blocks
- B. Input, orchestration, retrieval, safety, observability
- C. Training, inference, evaluation, deployment, monitoring, scaling, tuning
- D. Encoder, decoder, softmax, backprop, gradient, loss, epoch

22. What is a static embedding?

- A. A vector that changes every time a word appears in a new sentence
- B. A fixed vector for a token that doesn't yet know what sentence it's in
- C. The final output of the model
- D. A positional tag

23. How does a contextual embedding differ from a static embedding?

- A. It's smaller in size
- B. It gets nudged and updated based on surrounding words, so the same word can land in different meaning spots depending on context
- C. It's computed only once during training
- D. It ignores word order

24. Why are positional embeddings needed in a Transformer?

- A. Because attention alone can't tell word order, since tokens are processed in parallel
- B. Because words need a unique ID number
- C. Because they replace the need for tokenization
- D. Because they reduce compute cost

25. What are the three vectors every token produces in Q/K/V attention?

- A. Input, output, hidden
- B. Query, Key, Value
- C. Encoder, decoder, softmax
- D. Weight, bias, gradient

26. Why does multi-head attention use multiple heads instead of just one?

- A. To reduce the model size
- B. Because one attention pass can only focus on one kind of relationship at a time; multiple heads can each learn a different pattern
- C. To speed up tokenization
- D. To eliminate the need for feed-forward layers

27. What does the feed-forward layer do to each token's vector?

- A. Combines it with every other token's vector

B. Privately processes and refines what the token learned from attention, expanding then compressing the vector

C. Deletes positional information

D. Converts it back into text

28. What is a single "Transformer block" composed of?

A. Tokenization plus positional embedding only

B. Multi-head attention plus feed-forward refinement (with residual/normalization steps)

C. Query and Key only

D. Training and inference

29. Real GPT-style models chain how many Transformer blocks?

A. Always exactly 1

B. Dozens to well over a hundred

C. Exactly 7

D. Never more than 12

30. Whose visual series inspired the conceptual explanations in the "Understanding Transformers" deck?

A. Andrew Ng

B. 3Blue1Brown

C. Andrej Karpathy

D. Yann LeCun

Part D — The Evolution of "Attention Is All You Need"

31. What was the fixed-length bottleneck of the 2014 Seq2Seq model?

A. It required GPUs to train

B. The entire input sentence was squeezed into one fixed-size vector, losing meaning for long sentences

C. It couldn't handle short sentences

D. It had no decoder

32. What problem did Bahdanau Attention (2014) solve?

A. The quadratic cost of self-attention

B. The fixed-length vector bottleneck, by letting the decoder attend to every encoder hidden state

C. The vanishing gradient problem entirely

D. The need for tokenization

33. What limitation still remained after Bahdanau Attention?

A. It was too parallelizable

B. It was still fundamentally sequential (RNN-based), making training slow and hard to parallelize

- C. It couldn't be used for translation
- D. It required convolutional layers

34. How did Luong Attention (2015) differ from Bahdanau Attention?

- A. It removed attention entirely
- B. It generalized/simplified how attention scores are computed and introduced global vs. local attention windows
- C. It replaced RNNs with convolutions
- D. It introduced positional encodings

35. What did ConvS2S (2017) replace RNNs with?

- A. Transformers
- B. Stacked convolutions plus attention, allowing parallel computation
- C. Decision trees
- D. A single fixed vector

36. In "Attention Is All You Need" (2017), what did the Transformer drop entirely?

- A. Attention mechanisms
- B. Recurrence and convolution, relying only on multi-head self-attention and feed-forward layers
- C. Positional encodings
- D. Feed-forward layers

37. Which of the following is a limitation of the original Transformer architecture noted in the deck?

- A. It cannot be parallelized at all
- B. Self-attention scales quadratically ($O(n^2)$) with sequence length
- C. It has no need for large datasets
- D. It has unlimited context windows

38. What limitation did BERT (2018) address, and what new drawback did it introduce?

- A. Addressed generation tasks; introduced bidirectional bias
- B. Addressed bidirectional pretraining for strong transfer learning; introduced being encoder-only, so not naturally suited to text generation
- C. Addressed quadratic cost; introduced sparse attention
- D. Addressed vision tasks; introduced masked images

39. What limitation did Sparse / Longformer / Reformer models (2019-20) address?

- A. The unidirectional context problem
- B. The $O(n^2)$ cost of full attention, using sparse, local, or LSH attention, trading some global-context accuracy for efficiency
- C. The lack of positional encodings
- D. The need for convolutional layers

40. What made GPT-3 (2020) significant, and what drawback came with it?

- A. Significant for image processing; drawback was small model size
- B. Significant for emergent few-shot learning at scale; drawback was enormous compute/energy cost and hallucination risk
- C. Significant for encoder-only design; drawback was no generation capability
- D. Significant for reducing training data needs; drawback was slower inference

Part E — Scenario-Based: AI Application Architecture

41. A user sends a message that appears to be a jailbreak attempt trying to bypass the AI's safety policies. At which point in the pipeline should this be caught, and why?

- A. Observability layer, after the response is logged
- B. Input Layer, because it's the cheapest point to stop a bad query before it consumes any compute
- C. Orchestration layer, since it decides which model to call
- D. Safety & Quality layer, only after the model has generated a response

42. A support chatbot needs to answer questions using the company's internal product manuals rather than the model's general training knowledge. Which component should the team implement?

- A. Multi-Model Router
- B. RAG Pipeline (Vector DB) in the Retrieval & Tools layer
- C. Cost Tracker in the Observability layer
- D. Output Guardrails

43. A team wants simple FAQ queries answered by a fast, cheap model while complex reasoning tasks go to a frontier model like Claude or GPT-4, without hardcoding the choice per request. Which component solves this?

- A. Agent Orchestrator only
- B. Multi-Model Router
- C. Evaluation Framework
- D. Context & Session Memory

44. A generated response scores 40/100 on the Evaluation Framework, below the passing threshold. What should happen next according to the architecture?

- A. It is shown to the user with a disclaimer
- B. It is sent back to the Orchestrator for retry or refinement rather than shipped to the user
- C. It is permanently logged as a failure with no further action
- D. The Input Layer re-validates the original query

45. Leadership wants a monthly report on which users and models are driving the highest AI spend, plus where latency bottlenecks occur. Which layer supplies this data?

- A. Safety & Quality Layer
- B. Retrieval & Tools Layer
- C. Observability Layer (Cost Tracker, Latency & Performance, Eval Metrics Dashboard)
- D. Input Layer

Part F — Scenario-Based: A Field Guide to Transformers

46. In the sentence "I made a sweet Indian rice dish," the word "dish" should be understood to mean something closer to "biryani" rather than a generic dish. Which mechanism makes this possible?

- A. Static embeddings alone
- B. Positional embeddings alone
- C. Contextual embeddings shaped by attention over surrounding words like "sweet" and "Indian"
- D. The tokenizer's vocabulary size

47. A team is building a search/classification system that needs rich bidirectional understanding of a sentence (e.g., filling in a masked word) rather than generating new text. Which half of the Transformer should they use, and which model exemplifies it?

- A. Decoder only, like GPT
- B. Encoder only, like BERT
- C. Both encoder and decoder equally
- D. Neither — use only positional embeddings

48. A developer wants to build an autoregressive assistant that generates one word at a time, feeding each prediction back in as input for the next. Which architecture half fits, and which model is the classic example?

- A. Encoder only, like BERT
- B. Decoder only, like GPT
- C. Feed-forward network only
- D. Static embedding layer only

49. While debugging, an engineer swaps the word order in a sentence ("I made" → "made I") and notices the model's output changes even though the same words and static embeddings are used. What explains this sensitivity to order?

- A. The vocabulary size differs
- B. Positional embeddings, added after static embeddings, encode word order explicitly
- C. The softmax function changes
- D. The difference between training and inference

50. In the Query/Key/Value analogy, a professor writing an essay on quantum computing draws heavily from the student whose expertise best matches the essay's focus, and almost nothing from an unrelated student. Which part of the analogy does the "essay's focus" represent?

- A. Key
- B. Value**
- C. Query
- D. Softmax

Part G — Scenario-Based: Understanding Transformers (The 7-Step Pipeline)

51. A model needs to process the sentence "The cat sat" so that every token can look at every other token simultaneously (not one at a time like an RNN), yet still know that "The" comes before "cat." Which step ensures order is preserved under parallel processing?

- A. Static embeddings
- B. Positional embeddings**
- C. Feed-forward refinement
- D. Stacking N blocks

52. In "We sat by the river bank" vs "She deposited cash at the bank," the word "bank" ends up in different regions of the model's meaning space by the time attention finishes. Which pipeline stage is directly responsible for this shift?

- A. Tokenization
- B. Q/K/V attention producing contextual embeddings**
- C. Positional embeddings
- D. Static embeddings

53. A researcher observes that one attention head tracks grammatical structure while another tracks who is being referred to in a sentence, and their outputs are then combined. Which mechanism is being described?

- A. Feed-forward refinement
- B. Multi-head attention**
- C. Tokenization
- D. Positional embeddings

54. After attention lets tokens share context, each token is passed independently through a small neural network that expands then compresses its vector, with no communication between tokens at this step. Which stage is this?

- A. Multi-head attention
- B. Feed-forward refinement**
- C. Q/K/V attention
- D. Stacking N blocks

55. A model effectively "re-reads" a paragraph multiple times: the first pass gives the gist, later passes connect pronouns to names, and by the fifth pass subtle jokes are noticed. Which

concept does this re-reading analogy represent?

- A. Tokenization
- B. Static embeddings
- C. Stacking attention + feed-forward into N blocks
- D. Training vs. inference

Part H — Scenario-Based: The Evolution of "Attention Is All You Need"

56. A team in 2014 builds a translation system using two RNNs where the encoder compresses an entire sentence into one fixed-size vector before the decoder unfolds it. They notice translation quality drops sharply for long sentences. Which architecture are they using, and what is the core problem?

- A. Transformer; quadratic attention cost
- B. Seq2Seq; the fixed-length vector is an information bottleneck
- C. BERT; encoder-only limitation
- D. ConvS2S; limited receptive field

57. A team upgrades their Seq2Seq model so the decoder can dynamically weigh and "look back" at every encoder hidden state at each decoding step, solving the fixed-vector bottleneck — but training is still slow because the RNN backbone processes steps sequentially. Which innovation is this, and what limitation remains?

- A. Luong Attention; no remaining limitation
- B. Bahdanau Attention; still sequential, hard to parallelize on GPUs
- C. ConvS2S; limited receptive field
- D. Transformer; $O(n^2)$ cost

58. A team wants faster training by allowing computation across all sequence positions in parallel, so they replace RNNs with stacked convolutions plus attention. What architecture is this, and what new limitation does it introduce?

- A. Transformer; needs positional encodings
- B. ConvS2S; each convolution layer has a limited receptive field, requiring many stacked layers for long-range dependencies
- C. BERT; encoder-only
- D. GPT-3; huge compute cost

59. An organization needs to process very long documents (tens of thousands of tokens) but finds that standard self-attention becomes prohibitively expensive as sequence length grows. Which class of models specifically addresses this, and what do they trade off?

- A. BERT; trades bidirectionality for speed
- B. Sparse / Longformer / Reformer models; trade some global-context accuracy for reduced $O(n^2)$ cost
- C. GPT-3; trades data efficiency for scale

D. Vision Transformer; trades image data for text data

60. A company wants a model that generalizes to many tasks with just a few examples in the prompt, without task-specific fine-tuning, and is willing to accept very high compute and energy costs plus some risk of hallucination. Which model does this describe?

A. BERT (2018)

B. Transformer-XL (2019)

C. GPT-3 (2020), known for emergent few-shot learning at scale

D. Vision Transformer (2020)